

Supporting Information for “How Fragile Are LSTM Rainfall-Runoff Models? A Multi-Level Adversarial Robustness Assessment”

Y. Sun¹

¹College of Hydrology and Water Resources, Hohai University, Nanjing, China

Contents

1. Text S1: Model Training Configuration
2. Text S2: Basin Coverage and Missing Data
3. Text S3: Baseline Model Performance
4. Text S4: Static Attribute Ablation Details
5. Table S1: Model Training Hyperparameters
6. Table S2: Per-Experiment Basin Counts and Coverage
7. Table S3: Baseline (Clean) NSE Distribution Statistics
8. Table S4: Static Attribute Ablation Model Configuration

Text S1. Model Training Configuration

The CudaLSTM model was trained using the neuralhydrology framework (v1.12.0) with the configuration summarized in Table S1. We follow the temporal evaluation protocol of Kratzert et al. (2019): training period 1 October 1999 to 30 September 2008, validation period 1 October 1980 to 30 September 1989, and test period 1 October 1989 to 30 September 1999. All 531 CAMELS-US catchments are used in every period (temporal split). The model incorporates 27 static catchment attributes from the CAMELS dataset (Addor et al., 2017), covering topography (e.g., mean elevation, slope), soil (e.g., sand/clay fraction, depth to bedrock), vegetation (e.g., LAI, forest fraction), and climate characteristics (e.g., aridity, precipitation seasonality).

The model was trained with the AdamW optimizer using a stepped learning-rate schedule: 3×10^{-3} for epochs 1–9, 1×10^{-3} for epochs 10–19, and 5×10^{-4} for epochs 20–30. No dropout was applied. The best model was selected based on validation-period NSE (epoch 26, validation median NSE = 0.757).

Table 1 — Model training hyperparameters.

Parameter	Value
Architecture	CudaLSTM (single-layer)
Hidden units	128
Training epochs	30
Batch size	256
Sequence length	365 days
Loss function	NSE
Optimizer	AdamW
Learning rate	$3 \times 10^{-3} \rightarrow 1 \times 10^{-3} \rightarrow 5 \times 10^{-4}$
Dropout	0.0
Dynamic inputs	Precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure
Static attributes	27 CAMELS attributes (topography, soil, vegetation, climate)
Training period	01/10/1999–30/09/2008
Validation period	01/10/1980–30/09/1989
Test period	01/10/1989–30/09/1999
Random seed	159193

Text S2. Basin Coverage and Missing Data

The adversarial evaluation was conducted on all 531 CAMELS-US catchments. Of these, 524 yielded valid adversarial results; 7 catchments produced numerical failures (NaN in ΔNSE) attributed to numerical instability during gradient computation. These 7 catchments are excluded from analyses that involve ΔNSE statistics but included in counts of total experiment records.

Table S2 summarizes basin counts per experiment block.

Because we use a temporal split (all 531 catchments appear in every period), there is no basin-level in-sample vs. out-of-sample distinction. The adversarial evaluation is conducted on the held-out test *time period* (1989–1999), which is not seen during training.

Table 2 — Basin counts and coverage per experiment block.

Block	Description	Targeted	Valid	Coverage
Exp. 1	Attack comparison (5 methods, up to 4 ε)	531	524	98.7%
Exp. 2	Constraint ablation (3 constraints $\times$ 3 ε)	531	524	98.7%
Exp. 3	Targeted attacks (3 targets)	531	524	98.7%
Exp. 4	Causal trigger (4 windows)	531	524	98.7%
Exp. 5	C&W minimum perturbation	531	531	100%
Exp. 6	Static attribute ablation	531	524	98.7%
Total unique records (Exp. 1–5)			12,744	
Total unique catchments			531	

Text S3. Baseline Model Performance

Table S3 reports the distribution of baseline (clean, unperturbed) NSE values across the 531 evaluation catchments. The model achieves a median clean NSE of 0.599, which is lower than the ~ 0.74 reported by Kratzert et al. (2019). The discrepancy is primarily attributable to the different test period: our 1989–1999 period is climatically more challenging than the 2000–2005 period used in the original study.

Table 3 — Distribution of baseline (clean) NSE across 531 evaluation catchments.

Statistic	Value
Median NSE	0.599
Mean NSE	0.414
Q25 (25th percentile)	0.353
Q75 (75th percentile)	0.757
Fraction NSE > 0.5	61.8%
Fraction NSE > 0.7	33.5%
Fraction NSE < 0	10.9%

The adversarial vulnerability analysis in the main text (Section 3.7, Figure 6) explicitly accounts for baseline performance differences by stratifying results into perfor-

mance tiers. The key finding—that even well-modeled catchments (clean NSE > 0.7) exhibit a 3.9% catastrophic failure rate—demonstrates that adversarial vulnerability is not merely a consequence of poor baseline model performance.

Text S4. Static Attribute Ablation Details

The static attribute ablation experiment (Section 3.8 of the main text) compares two models that differ only in their use of static catchment attributes. The ablation model configuration is shown in Table S4.

Table 4 — Static attribute ablation: model configurations. All parameters except static attributes are identical between the two models.

Parameter	Primary model	Ablation model
Static attributes	27 CAMELS attributes	None
Test median NSE	0.599	0.421
Config file	<code>train_final_with_static.yml</code>	<code>train_final_no_static.yml</code>

The ablation model was evaluated under three attack methods at $\varepsilon = 0.1$ (L_p constraint, untargeted): Auto-PGD, FGSM, and Gaussian noise. Results for 524 matched catchments (same 7 excluded due to NaN) are compared using the Wilcoxon signed-rank test (paired, non-parametric). The primary model shows significantly less adversarial degradation across all gradient-based attacks (Auto-PGD: $p = 1.8 \times 10^{-30}$; FGSM: $p < 10^{-10}$). The Gaussian noise baseline shows no significant difference between models (Δ NSE of -0.030 vs. -0.032), confirming that the robustness improvement is specific to adversarial (gradient-exploiting) perturbations, not a general reduction in sensitivity.

References

- Addor, N., Newman, A. J., Mizukami, N., & Clark, M. P. (2017). The CAMELS data set: catchment attributes and meteorology for large-sample studies. *Hydrology and Earth System Sciences*, 21(10), 5293–5313.
- Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S., & Nearing, G. (2019). Towards learning universal, regional, and local hydrological behaviors

via machine learning applied to large-sample datasets. *Hydrology and Earth
System Sciences*, 23(12), 5089–5110.